equally unfortunate star and spectacularly rips itself apart. Because all white dwarf stars are fairly simple objects,[fn9] and because the explosion is governed by physics that we feel we have a somewhat decent handle on, Type Ia's were considered for a time to be good standard candles—the explosions all looked pretty similar. But it was later found that they're better described as standard*izable*, in the same way Cepheid variables are. If you can measure how the explosion peaks and dims, you can get a good sense for the total amount of energy put out by the explosion, and thus an idea of how bright it really is.

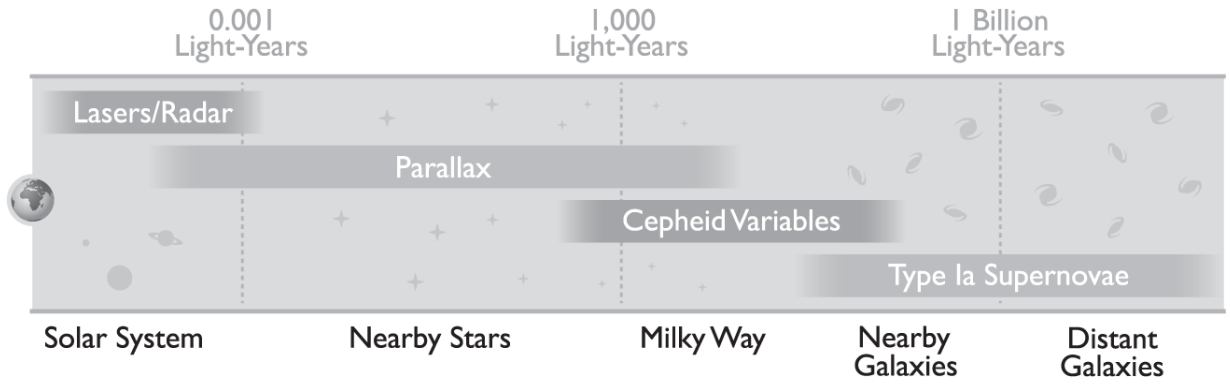


Figure 16: Cosmic distance ladder. For bodies within the Solar System, we can use lasers or radar (in addition to relationships between orbital times and distances) to measure distances. Distances to nearby stars can be measured with parallax, and Cepheid variable stars can help us determine distances within the Milky Way and some nearby galaxies. For more distant sources, we can use Type Ia supernovae.

STAR LIGHT THERMONUCLEAR BRIGHT

But this book is about destruction, and I would be remiss if I glossed over Type Ia supernovae as just "a kind of exploding star." A white dwarf star, the kind of star our Sun is fated to eventually become, is itself a marvel of stellar evolution. And when one explodes, it does so by undergoing an all-out full-body thermonuclear detonation that outshines its entire galaxy.

If you are any kind of star, no matter what stage you are at in your life cycle, your existence depends on a careful balancing act between the pressure produced in your core and the gravity of the material you're made of. (We call this "hydrostatic equilibrium" but it really just comes down to the idea that the push out has to be equal to the pull in for a star to neither blow up nor collapse.) Most of the time, a star creates outward pressure by doing fusion reactions in its core—pressing nuclei together so tightly that